Lab 1: Use-Case Flow - UC-03 Place Order

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Field	Detail
Use case	UC-03 Place Order
Primary actor	Customer
Secondary actors	Payment Gateway, Receipt Printer
Preconditions	Kiosk is on the welcome screen, menu and prices are loaded, printer has paper.
Postconditions (success)	Order is stored as PAID with an order number and a receipt is printed.
Trigger	Customer taps "Start Order".
Includes / extends	«include» UC-04 Pay by Credit Card, «include» UC-05 Print Receipt; UC-07 Retry Payment «extend» UC-04

Main Success Scenario

1. Customer taps **Start Order**. System displays the menu (Espresso, Americano, Latte) with prices.
2. Customer selects a coffee type (UC-01). System highlights it and shows the size options.
3. Customer selects Small or Large (UC-02). System shows the order summary and total price.
4. Customer taps **Pay by Card**. System starts UC-04 and asks the customer to insert or tap a credit card.
5. Customer presents the credit card to the card reader.
6. System sends an encrypted authorization request to the **Payment Gateway**.
7. Payment Gateway approves the payment.
8. System stores the order as PAID and generates an order number.
9. System starts UC-05 and sends the receipt to the **Receipt Printer**, which prints it.
10. System shows "Order #N confirmed - collect your receipt" and returns to the welcome screen after 10 s. Use case ends successfully.

Alternate Flows

7a. Payment declined (UC-07 Retry Payment extends UC-04)

- 7a1. System displays a **Payment declined** message; no order is stored.
- 7a2. Customer is prompted to try another card or cancel.
- 7a3. If the customer retries, the flow resumes at step 5.
- 7a4. If payment fails again (after 2 attempts), the order is **cancelled**, the customer is notified, and the kiosk returns to the welcome screen.

6a. Payment Gateway does not respond within 15 s

- 6a1. System shows "Payment service unavailable - you have not been charged".
- 6a2. System offers one retry; if it also times out the order is cancelled.

9a. Printer out of paper or offline

- 9a1. System shows the order number in large text on screen so the customer can still collect the drink.
- 9a2. System logs the printer fault and alerts the Administrator. Use case ends successfully.

1-4a. Customer cancels before paying: at any of steps 1-4 the customer taps Cancel; the system discards the selection and returns to the welcome screen.